Ahnaf Shahriar

Email | LinkedIn | Github

Education

University of Waterloo Waterloo, ON
Bachelor of Applied Science in Computer Engineering *Sept. 2021 – Apr. 2026*

- Recipient of Richard & Elizabeth Madter Entrance Scholarship and President's Scholarship of Distinction
- **Relevant Courses**: Computer Architecture(Verilog), Computer Networks, Systems Programming and Concurrency, Embedded Microprocessing Systems, Analog Control Systems, Compilers(Java), Numerical Methods

Experience

Wireless Firmware Developer Jan. 2025 – Apr. 2025
ON Semiconductor Corporation *Waterloo, ON*

- **SDK Development**: Integrated *I3C support* and *BLE* profile services into sample code.
- **Wireless Connectivity**: Developed and integrated cutting-edge *LE Audio, HFP, and A2DP* functionality into an all-in-one Application.
- **Wireless Manager**: Modularized and implemented a state machine for power efficiency by *25%*
- **Test Automation**: Automated SDK test cases, reducing manual testing efforts by up to *80%*

IC Design and Verification Intern May 2023 – Aug. 2023, Jan. 2024 – May 2024
NXP Semiconductors Canada *Kanata, ON*

- **IP Design**: Designed multiple IP blocks including *proprietery ECC IP* for dataplane processing chips.
- **Timing Analysis**: Spearheaded critical path improvements in IP block to increase Clock Frequency by *50%*.
- **Functional Testing**: Designed brand new End-to-End functional tests in simulating *10GBps+* traffic for IP.
- **Unit Test Planning**: Created Simulation scenarios for testing High speed Dataplane Processing features.

Embedded Software Engineering Intern Sept. 2022 – Dec. 2022
Synapse Product Development *Seattle, WA*

- **Prototyping**: Leveraged Zephyr RTOS to create a proof of concept on *NRF52 BLE* device.
- **Python APIs**: Developed company specific lab automation software for equipment from *Agilent, Keysight, NI, Tektronik.*
- **Automation**: Streamlined testing and in house procedures using *Python* and *Bash.*
- **Driver Development**: Designed and implemented *drivers* of automated PCB testing Device*(I2C, UART)*

Firmware developer Jan. 2022 – April 2022
Ford Motor Company of Canada *Remote*

- **Unity/Cmock Test framework**: Lead developer for optimization for unit testing, achieving up to *30% faster* runtime while using *50%* less manually written test cases.
- **Automation**: Improved *Jenkins CI/CD* pipelines to support unit testing automation using *Python* for Linux server.
- **Embedded Trace Debugging**: Developed Parsing tools to debug through CAN and Serial message parsing of *10 million+* lines.
- **Automotive Design**: Maintained *AUTOSAR* standard design with *ISO26262 safety design* using *Davinci Configurator.*

Projects

LC-3 Emulator: A C emulator for an educational ISA. Improves by 25% on research paper using *Python* data logging.
Real Time Executable: A RTOS implementation in STM32 capable of Pre-emptive task switching and its own Malloc
Stereo System: An embedded C implementation of a stereo playback system. Created with Quartus on Artix FPGA.
VHDL Compiler: A Java Compiler for creating combinational VHDL circuits. Using a boolean intermediate representation

Technical Skills

Languages: C/C++, Java, Python, Tcl, Bash scripting, ASM, VHDL, SystemVerilog/Verilog
Tools: Keil, Quartus, Git, Linux, Qemu, LLDB/GDB , Docker, WireShark, UVM, Matlab
Hardware: Oscilloscopes, Logic Analyzer, Multimeters, Spectrum Analyzer
Protocols: TCP/IP, JTAG, Serial, Ethernet, CAN/CAN-FD, LIN